

4. Core Functionalities Development

ComicCraft Project Documentation

Purpose: Explains the five core processing functions and how they work together to transform a prompt into a downloadable comic.

4.1 Structured Comic Outline

Function: `generate_outline()`

File: `gemini_flash.py`

The documented function uses Gemini Flash to generate a structured 5-panel comic outline from the user's story prompt. Each panel contains a panel number, title, scene description, and image-generation prompt.

```
generate_outline(story_prompt)
→ 5 panel dictionaries
→ panel number + title + scene description + image prompt
```

4.2 Story Narration and Dialogue

Function: `generate_story()`

File: `gemini_pro.py`

Gemini Pro expands the panel outline into a full comic-style story containing engaging narration and character dialogues.

```
generate_story(outline)
→ formatted comic story
→ narration + character dialogue
```

4.3 Comic Illustration Generation

Function: `generate_image()`

File: `image_generator.py`

Stable Diffusion creates comic-style images from image prompts. The generated image is saved under `static/panels`, and the function returns the file path.

```
generate_image(image_prompt)
→ Stable Diffusion
→ static/panels/
```

4.4 Layout Construction

Function: `build_comic_layout()`

File: `layout_builder.py`

This function combines the generated images with the corresponding panel story. The resulting layout contains panel number, image path, and text.

4.5 PDF Export

Function: `save_pdf()`

File: `exporters.py`

FPDF compiles the comic into a multi-page PDF. Each panel's image and narration are placed on separate pages, and the PDF is stored in `static/exports`.

4.6 Core Pipeline

```
generate_outline()  
↓  
generate_story()  
↓  
generate_image() for each panel  
↓  
build_comic_layout()  
↓  
save_pdf()
```